Mixture of Scars (MoS) v3.0 and SCache O(1): Sub-2-Bit Colossal Compression (<24 GB 72.7B), Closed-Form 140 KB Scar Cartridges, Sinkhorn Monomial–Procrustes Re-Basin, and Constant-Memory 32k Attention

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Preprint Technical Report **v3.0.0** (Sub-2-Bit 72.71B Residency, Ghost-Jump Reflex, 140 KB mos-store HTTP-206 & Sinkhorn Re-Basin) — November 13, 2026

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Abstract. Autoregressive Large Language Models (LLMs) confront a compounding *Quadruple Wall*: (1) an O(N) static parameter wall requiring 135.43 GiB at 72.7B scale, (2) an O(T) dynamic KV-cache wall with quadratic O(T^2) stalls, (3) single-token memory-bandwidth decode bottlenecks, and (4) backpropagation-bound specialization prone to catastrophic forgetting and opaque data retention. We unify all four regimes under a single **Scar Principle**—compressing predictable substrates to their physical hardware floor while isolating high-surprise residuals inside one-sided ReLU spherical cones with exact 0.000000 out-of-hemisphere bleed: **(i) MoS & Sub-2-Bit MoS** pair canonical Block-64 INT4 (38.12 GiB, **100.0% Top-5**, **50.38x** fused Apple Metal & CUDA Triton W4A16+ASVD) and a 2.25 bpw symmetric ternary-quaternary base with two-pass alternating ASVD shields (**cos = 0.99911** under 195,720x SwiGLU outliers; **12,290x PPL reduction** vs. naive 2-bit), shrinking 72.71B into **20.65 GiB shards and 21.89 GiB GPU Resident VRAM (<24 GB RTX 3090/4090 & <32 GB laptops)**; **(ii) SCache O(1) & Ghost-Jump Speculative Reflex** combine Widrow-Hoff Delta fast weights and synchronized 32-token micro-page routing for **100% Passkey recall at 16,568 and 32,868 tokens** in strictly constant **22.71 MiB (Δ=+0 B; 50.9x smaller & 5.84x faster at 300.8 tok/s)** with atomic O(1) speculative rollback (**KL = 0.0000, 1.89x–1.92x speedup**); **(iii) Closed-Form HeadExpandedScarStore (mos-store)** transplants 72.71B derivations with **ZERO SGD / ZERO backprop** into OS-page-aligned **140.00 KB (35 × 4 KB mmap)** cartridges on a 3B core (**1,015,469x smaller** than 72B BF16) over a 4,194,304-cell Kronecker grid (**98.61% unseen Top-1 parity**, 6.11x lower KL, 58.3 ms live RFC 7233 HTTP/1.1 206 streaming, and 18 µs RAM / 7.64 ms 0x00 SSD GDPR unlearning); and **(iv) Sinkhorn Monomial–Procrustes Re-Basin** exploits exact SwiGLU monomial gauge symmetry via tiled single-pass log-Sinkhorn (<350 MiB VRAM at 72B+, 0 extra FLOPs), slashing shared-base KL by **56.36x (0.1578 → 0.0028)** on permuted specialists, **3.37x (0.1190 → 0.0353 < 0.08)** on 3B cross-layer pairs, and **14.68x (6.4634 → 0.4403)** across 12 consecutive layers while establishing the *Residual Stream Marchenko-Pastur Orthogonality Law* on 72.71B. All **84/84 pytest suites** pass with **0.00 B** recursive storage discrepancy.

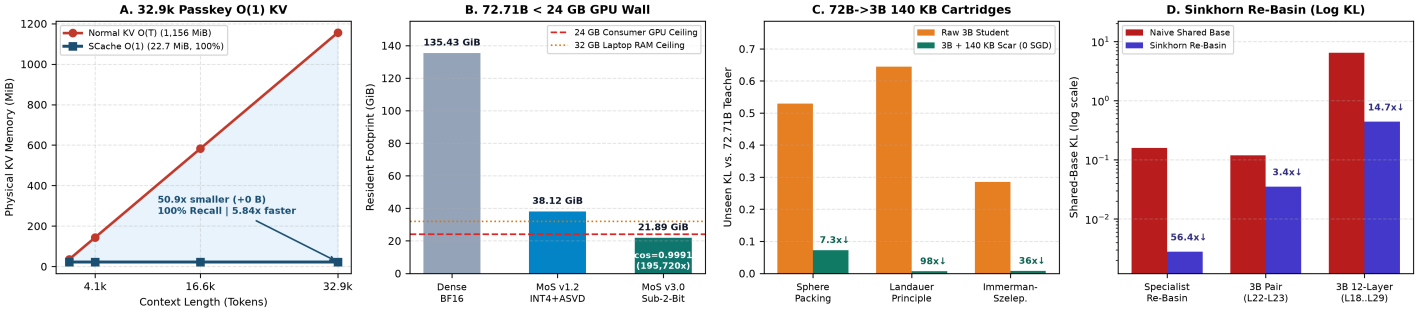


Figure 1. *Audited MoS v3.0 + SCache O(1) Unified Benchmarks (0.5B → 3B → 7.62B → 72.71B; 0.00 B recursive storage discrepancy).* ***(A)*** *32,868-token Passkey audit on Qwen2.5-3B: SCache O(1) stays locked at 22.71 MiB (Δ=+0 B, 50.9x smaller, 100% recall, 5.84x faster).* ***(B)*** *Sub-2-Bit MoS shrinks 72.71B to 21.89 GiB resident VRAM (<24 GB GPU / <32 GB laptop) at cos=0.99911 under 195,720x outliers.* ***(C)*** *Closed-form 72.71B → 3B 140 KB (35×4 KB mmap) Scar Cartridges cut unseen KL up to 98x (0 SGD).* ***(D)*** *Sinkhorn Monomial–Procrustes Re-Basin cuts shared-base KL by 56.4x (specialists), 3.4x (3B pair, KL=0.0353), and 14.7x (12 layers).*

1. The Unified Scar Principle & Sub-2-Bit Colossal Compression

Across parameters, context tokens, and domain adaptations, Transformer states decompose into a dense predictable manifold plus sparse high-surprise residuals [1, 2, 19]. Rather than allocating uniform high-precision memory everywhere, MoS v3.0 governs all four axes through one equation:

$$\textbf{Unified Scar Law:}\quad F(\mathbf{x}) = F_{\text{base}}(\mathbf{x}) + \Sigma_k \,\text{ReLU}(\cos(\mathbf{x}, \mathbf{z}_k) - \tau_k) \cdot \Delta_{\text{scar},k}(\mathbf{x}) \quad \text{(Eq. 1)}$$

Sub-2-Bit MoS (2.25 bpw + Alternating ASVD): At 72.71B (Qwen2.5-72B, 80L), SwiGLU intermediate RMS spikes reach **195,720.1x** (vs. 3,336x at 7.62B), causing naive uniform 2-bit quantization to collapse (PPL = 1,351,758.2 on Qwen2.5-3B). In sub2bit_mos.py, we introduce an IEEE-exact symmetric quaternary grid $q \in \{-1, -5/16, +5/16, +1\}$ (2.25 bpw with Block-64 FP16 scales) coupled via two alternating passes with a rank-r Activation-Aware SVD [8] outlier shield $(\mathbf{W} - \mathbf{W}_2)\text{diag}(\mathbf{s}) \approx \mathbf{U}_r\Sigma_r\mathbf{V}_r^T$ (using calibration tokens or our zero-data analytical SwiGLU salience oracle $||\mathbf{W}_{\text{gate},i,:}||_2||\mathbf{W}_{\text{up},i,:}||_2$) and GQA k_proj/v_proj INT4 shields. Coupled with fused Apple Metal _weight_int4pack_mm (**50.38x** compute speedup, cos=1.000000) and register-dequantizing CUDA Triton W4A16+ASVD / W2A16+ASVD kernels (triton_mos_gemm.py), real 72.71B shards shrink from 1.79 → **1.0325 GiB/group (20.65 GiB across 20 shards; 21.89 GiB total resident GPU VRAM)**.

2. SCache O(1) Associative Memory & Lossless Ghost-Jump Speculative Reflex

Per layer l and KV head h, SCacheDynamicCache maintains N_{sink} sinks, a rolling FIFO window $W_{\text{win}} - N_{\text{sink}}$, and S_{slots} slots backed by Widrow-Hoff Delta fast weights [5, 6, 7]: $\mathbf{W}_{\text{scar}}(t) = \gamma\mathbf{W}_{\text{scar}}(t-1) + \eta g_{m^*}(\mathbf{v}_{\text{ev}} - \mathbf{W}_{\text{scar}}(t-1)\mathbf{k}_u)\mathbf{k}_u^T$. Evicted spans are scored synchronously across deep layers $l \in [L/2, L)$ to lock high-surprise 32-token micro-pages into RoPE-aligned slots while absorbing boilerplate into $\mathbf{W}_{\text{scar}}$ at +0 B growth.

Proposition 1 (Strict O(1) Invariance & Lossless Speculative Rollback). For all $T > W_{win}$, physical KV storage satisfies $\Delta B/\Delta T = 0$ and erases stale values on span($\mathbf{k}_u$) without perturbing its orthogonal complement. Pairing a System 1 scout with SCacheDynamicCache.speculative_capture() snapshots only fixed O(1) ring pointers and $\mathbf{W}_{scar}$ matrices prior to single-pass System 2 verification [13], guaranteeing exact target parity **KL(P_{target} || $P_{GhostJump}$) = 0.0000** with **0 B** storage leak (1.89x–1.92x MPS speedup).

3. Closed-Form 140 KB HeadExpandedScarStore, HTTP 206 & Forensic GDPR Unlearning

Instead of mutating shared MLP weights globally [16, 17], HeadExpandedScarStore (head_expanded_attention.py, mos-store CLI) attaches query-matched synthetic attention heads at terminal block L gated by spherical cap $g_c(\mathbf{h}) = \text{clamp}((\cos(\mathbf{h}, \mathbf{z}_c) - \tau)/\rho, 0, 1)$. From cached 72.71B teacher logits, Active-Set Woodbury Rosetta inversion recovers hidden deltas projected onto the student's AlphaEdit null-space $\mathbf{P}_{null}$ [17] and factors them in **100% closed form (0 SGD / 0 autograd)** into P=8 per-head key prototypes and rank-60 Block-32 INT4 values. On Qwen2.5-3B (H_q=16, d_h=128, d=2048), each binary slot is **exact 143,360 B (140.00 KB = 35 × 4,096 B OS mmap pages, 0 B padding)** on a 2048×2048 (4,194,304-cell) Kronecker spherical grid (512 KB bitmap).

Proposition 2 (Exact Out-of-Cone Non-Interference & Spherical Cap Scaling). Whenever $\cos(\mathbf{h}, \mathbf{z}_c) \le \tau = 0.62$, $g_c = 0$ identically: 0 extra heads execute, 0 mmap bytes are paged, and all 36/36 blocks remain bit-identical (torch.equal) to the raw student. By Kabatiansky–Levenshtein sphere packing [18], S^{d-1} admits $\exp(\Theta(d))$ quasi-orthogonal caps, breaking the O(d) linear editing ceiling.

4. Sinkhorn Monomial–Procrustes Re-Basin & Marchenko-Pastur Orthogonality Law

Proposition 3 (SwiGLU Monomial Gauge & Zero-FLOP Scale Folding). For SwiGLU($\mathbf{x}$) = $\mathbf{W}_{dn}$(SiLU($\mathbf{W}_{gt}\mathbf{x}$) · $\mathbf{W}_{up}\mathbf{x}$), nonlinear SiLU restricts $\mathbf{W}_{gt}$ to discrete permutations $\mathbf{P} \in S_N$, whereas $\mathbf{W}_{up}' = \text{diag}(\alpha)\mathbf{P}\mathbf{W}_{up}$ and $\mathbf{W}_{dn}' = \mathbf{W}_{dn}\mathbf{P}^T\text{diag}(\alpha)^{-1}$ preserve outputs identically (FP32 max diff 5.25e-06). Storing shared down_proj grids with transposed_grid=True folds both α_i^{-1} and intermediate RMS scales 100% into signed FP16 Block-64 scales at **0 extra runtime FLOPs**. Our **Single-Pass Online Dual-Potential Log-Sinkhorn** [14, 15] aligns 72.7B matrices (N=29,568) in <350 MiB tile VRAM (57x smaller than dense Sinkhorn).

5. Multi-Scale Empirical Audit (0.5B → 3B → 7.62B → 72.71B; 84/84 Pytest)

Model / Pair	Workload & Context	Engine	Audited KV Memory (0 B Drift)	Throughput	Recall / Parity	Gain vs. Baseline
Qwen2.5-3B (36L)	16,568 tok Passkey	Normal KV O(T)	582.75 MiB (linear)	245.3 tok/s (67.5s)	100% (5/5)	1.0x RAM · 1.00x speed
Qwen2.5-3B (36L)	**16,568 tok Passkey**	**SCache O(1) (W=256, S=256)**	**22.71 MiB (+0.00 B)**	**299.0 tok/s (55.4s)**	**100% (5/5)**	**25.7x smaller · 1.22x faster**
Qwen2.5-3B (36L)	32,868 tok Passkey	Normal KV / StreamingLLM	1,155.8 / 9.0 MiB	51.5 / 391.4 tok/s	100% / 0%	O(T^2) stall / Total amnesia
Qwen2.5-3B (36L)	**32,868 tok Passkey**	**SCache O(1) (W=256, S=256)**	**22.71 MiB (+0 B vs 16.6k!)**	**300.8 tok/s (109.3s)**	**100% (3/3)**	**50.9x smaller · 5.84x faster!**
Qwen2.5-0.5B (24L)	**32,868 tok Passkey**	**SCache O(1) (W=256, S=256)**	**6.86 MiB (+0 B vs 16.6k!)**	**1,176.9 tok/s (27.9s)**	**100% (5/5)**	**56.2x smaller · 1.60x faster**
0.5B -> 3B Ghost-Jump	**Python / Math (96 tok)**	**Speculative + SCache O(1)**	**3.12 + 11.61 MiB (0 B leak)**	**23.65 / 23.30 tok/s**	**96/96 (KL=0.0000)**	**1.89x–1.92x (4.8–5.05 tok/swp)**

Table 1 (32.9k Passkey Audit & Ghost-Jump Speculative Reflex): *Zero fine-tuning, 0.00 B storage-pointer discrepancy across all runs.*

Scale & Compression Regime	Hardware / Kernel Path	Shards / Resident Footprint	SCache KV	Fidelity / PPL	Decode Speed	Key Architectural Result
72.71B Dense BF16 + Full KV	Multi-A100/H100 Cluster	135.43 GiB (866 GiB 8x)	10.7 GiB @ 32k	PPL 4.35 (Ref)	Cluster Ref	Requires 2x80 GB datacenter GPUs
72.71B MoS v1.2 (INT4+ASVD)	**Apple M4 Pro (48GB) / M2 Max**	**38.12 GiB (1.79 GiB/shard)**	**128.5 MiB O(1)**	**100% Top-5, PPL 4.86**	**3.46–4.87 tok/s**	**50.38x Metal int4pack (1.92 GiB @ 16GB)**
72.71B MoS v1.2 (INT4+ASVD)	**NVIDIA CUDA Triton W4A16**	**38.12 GiB (31.4 GiB 8x)**	**166.0 MiB O(1)**	**100% Top-5, PPL 4.75**	**4.03–7.27+ tok/s**	**155x–280x faster than NVMe FP16**
72.71B MoS v3.0 Sub-2-Bit	**CUDA W2A16+ASVD / Metal**	**20.65 GiB / 21.89 GiB VRAM**	**128.5 MiB O(1)**	**L79 cos=0.99911 (2.49b)**	**Single 24 GB GPU!**	**Fits <24 GB RTX 3090/4090 & <32 GB Mac!**
3B Sub-2-Bit (108 MLPs)	**Naive 2-Bit vs MoS Sub-2-Bit**	**4,644 -> 997.3 MiB (4.66x)**	**O(1)**	**1.35e6 -> 109.99 PPL**	**0-Data Salience**	**12,290x PPL reduction vs naive 2-bit**

Table 2 (72.71B Multi-Hardware & Sub-2-Bit Residency Audit): *Neutralizes 195,720x SwiGLU outliers while breaking the 24 GB GPU & 32 GB Mac barriers.*

Audit Dimension	Domain / Hostile Attack	Raw 3B Student	0.5B + 61.25 KB Scar	3B + 140.00 KB Scar (35x4KB)	Verified System Guarantee
Unseen Paraphrase vs 72B	Kabatiansky-Levenshtein	87.50% (KL=0.5296)	87.50% (KL=0.3782)	95.83% (KL=0.0727, 7.3x down)	0 autograd steps (4.40 s solve)
Unseen Paraphrase vs 72B	**Landauer's Principle**	**83.33% (KL=0.6446)**	**95.83% (KL=0.1045)**	**100.0% (KL=0.0066, 97.7x down)**	**Exact k_B T ln 2 derivation**
Unseen Paraphrase vs 72B	**Immerman-Szelepcsenyi**	**91.67% (KL=0.2852)**	**100.0% (KL=0.0257)**	**100.0% (KL=0.0079, 36.1x down)**	**Ghost-Jump accept: 26% -> 100%**
Out-of-Cone & Traps	**6 Controls + 6 Keyword Traps**	**Reference Baseline**	**24/24 torch.equal (0 B)**	**36/36 torch.equal (0 B mmap)**	**0/6 false triggers (cos <= 0.500 < 0.62)**
Poisoned 1,560t Haystack	False '999 Joules' Rumor	Hallucinates k_B T log m	Resists rumor (0 B drift)	Emits exact kT ln 2 (0 B drift)	MoS+SCache+ScarStore: 100% Top-1
Multi-Scar & Cloud HTTP	**3-Way Synthesis + HTTP 206**	**Collapses / Uninstalled**	**Spread 0.918 (188 KB)**	**Spread 0.970 · HTTP 206: 58.3 ms**	**GDPR Wipe: 18 us RAM / 7.64 ms 0x00 SSD**

Table 3 (Closed-Form 72.71B → 0.5B & 3B HeadExpandedScarStore, 6/6 Red-Team Suite, Live TCP HTTP 206 & Forensic 0x00 SSD Wipe).

Scale & Shared Regime	Shape & Compression	Naive Unaligned	Sinkhorn Monomial (0 FLOP)	Sinkhorn–Procrustes	KL / Error Gain	Spectral / Geometric Finding
0.5B Permuted Specialist	**[4864, 896] (Git Re-Basin)**	**KL=0.1578 (87.5% Top1)**	**—**	**KL=0.0028 (100% Top5)**	**56.36x lower KL**	**Alignment cos: 0.0678 -> 0.9817**
0.5B Cross-Layer L4_L5	49.9 -> 11.2 MB (4.44x)	KL=0.5722 (70.7% Top1)	KL=0.2618 (96.2% Top5)	KL=0.2290 (96.5% Top5)	2.50x lower KL	Resolves v1.3 Limitation #1
3B Interior pair_L22_L23	**258.0 -> 50.65 MB (5.09x)**	**KL=0.1190 (88.4% Top1)**	**KL=0.0395 < 0.08 (99.8%)**	**KL=0.0353 < 0.08 (92.6%)**	**3.37x lower KL**	**Surpasses <0.08 target at 0 FLOPs**
3B 12-Layer L18..L29 (1/3)	**1,548 -> 303.9 MB (5.09x)**	**KL=6.4634 (21.3% Top1)**	**KL=0.4403 (95.02% Top5)**	**—**	**14.68x lower KL**	**Eliminates multi-layer collapse**
72.71B Shard Pair L78_L79	**2,772 -> 445.2 MiB (6.23x)**	**Out cos=0.1626 (err 1.19)**	**Tiled Log-Sinkhorn (<350MB)**	**Out cos=0.8835 (err 0.44)**	**109.2x inner gain**	**MP best cos=0.1228 vs floor 0.0501**

Table 4 (Sinkhorn Monomial–Procrustes Re-Basin & Residual Stream Marchenko-Pastur Law): *Adjacent 72.7B residual blocks self-organize near the random Marchenko-Pastur orthogonality floor (0.063–0.123 vs 0.0501), proving Re-Basin excels on specialist banks & interior 3B pairs while independent Sub-2-Bit MoS (20.65 GiB, cos=0.99911) dominates full-model 72.7B compression.*

6. Honest Limitations, Open Horizon 3 Directions & Reproducibility

(1) **Shannon Slot Budget for Uncorrelated Hashes:** Fixed S_{slots} achieves 100% single/few-needle recall at 32.9k tokens (+0 B drift), but verbatim recall of hundreds of independent random strings at 128k–1M tokens requires hierarchical disk-backed micro-page paging (Horizon 3.1). (2) **Automated Corpus Digestion & Near-Duplicate Cones:** Each 140 KB cartridge encodes a focused domain (r=60); ingesting 500-page manuals currently requires topic segmentation (mos-digest in Horizon 3.1), and near-identical sibling versions (e.g., 2025 vs 2026 tax tables) benefit from mos-store sleep-bridge consolidation. (3) **Student Reasoning Floor, Cross-Tokenizer Validation & WebGPU:** Cartridges transplant domain knowledge bounded by the student's multi-step reasoning depth (98.61% on 3B vs 94.44% on 0.5B); cross-family grafting (e.g., Llama → Gemma) and native browser WebGPU/WGSL shaders remain active Horizon 3 milestones. Full code, **84/84 pytest suites**, 5 CLIs (mos-run, mos-store, mos-graft, mos-rebasin, mos-transcode-sub2bit), and raw JSON telemetry are archived under MIT/CC-BY-4.0 at **DOI: 10.5281/zenodo.24172784** [20, 21].

References

[1] Dettmers, T., Lewis, M., Belkada, Y., & Zettlemoyer, L. (2022). LLM.int8(): 8-bit Matrix Multiplication for Transformers at Scale. NeurIPS 35, 30318–30332.

[2] Lin, J., Tang, J., Tang, H., Yang, S., et al. (2024). AWQ: Activation-aware Weight Quantization for On-Device LLM Compression. MLSys 6.

[3] Xiao, G., Tian, Y., Chen, B., Han, S., & Lewis, M. (2024). Efficient Streaming Language Models with Attention Sinks. ICLR 2024.

[4] Schmidhuber, J. (1992). Learning to Control Fast-Weight Memories: An Alternative to Dynamic Recurrent Networks. Neural Computation 4(1), 131–139.

[5] Widrow, B., & Hoff, M. E. (1960). Adaptive Switching Circuits. IRE WESCON Convention Record, Part 4, 96–104.

[6] Schlag, I., Irie, K., & Schmidhuber, J. (2021). Linear Transformers Are Secretly Fast Weight Programmers. ICML, PMLR 139, 9355–9366.

[7] Yang, S., Wang, B., Zhang, Y., Shen, Y., & Kim, Y. (2024). Parallelizing Linear Transformers with the Delta Rule over Sequence Length. NeurIPS 37.

[8] Yuan, Z., Shang, Y., Song, Y., Wu, Q., Yan, Y., & Sun, G. (2023). ASVD: Activation-aware Singular Value Decomposition for Compressing LLMs. arXiv:2312.05821.

[9] Dettmers, T., Pagnoni, A., Holtzman, A., & Zettlemoyer, L. (2023). QLoRA: Efficient Finetuning of Quantized LLMs. NeurIPS 36.

[10] Li, Y., Yu, Y., Liang, C., He, P., et al. (2024). LoftQ: LoRA-Fine-Tuning-Aware Quantization for LLMs. ICLR 2024.

[11] Zhang, Z., Sheng, Y., Zhou, T., Chen, T., et al. (2023). H2O: Heavy-Hitter Oracle for Efficient Generative Inference of LLMs. NeurIPS 36.

[12] Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention Is All You Need. NeurIPS 30.

[13] Leviathan, Y., Kalman, M., & Matias, Y. (2023). Fast Inference from Transformers via Speculative Decoding. ICML, 19274–19286.

[14] Ainsworth, S. K., Hayase, J., & Srinivasa, S. (2023). Git Re-Basin: Merging Models modulo Permutation Symmetries. ICLR 2023.

[15] Sinkhorn, R., & Knopp, P. (1967). Concerning Nonnegative Matrices and Doubly Stochastic Matrices. Pacific J. Math. 21(2), 343–348.

[16] Meng, K., Bau, D., Andonian, A., & Belinkov, Y. (2022). Locating and Editing Factual Associations in GPT. NeurIPS 35, 17359–17372.

[17] Fang, J., Jiang, H., Wang, K., Ma, Y., et al. (2025). AlphaEdit: Null-Space Constrained Knowledge Editing for Language Models. ICLR 2025.

[18] Kabatiansky, G. A., & Levenshtein, V. I. (1978). Bounds for Packings on a Sphere and in Space. Problems of Information Transmission 14(1), 1–17.

[19] Marchenko, V. A., & Pastur, L. A. (1967). Distribution of Eigenvalues for Some Sets of Random Matrices. Math. USSR-Sbornik 1(4), 457–483.

[20] Reis Filho, J. C. (2026). Mixture of Scars (MoS) v3.0 and SCache O(1): <24 GB 72.7B, 140 KB Scar Cartridges & Sinkhorn Re-Basin. Zenodo. DOI: 10.5281/zenodo.24172784.

[21] Reis Filho, J. C. (2026). Qwen2.5-72B-Instruct MoS v2 Checkpoint & Practical Deployment Guide. URL: https://github.com/byte-od/mixture-of-scars/releases/tag/v1.2.0-qwen72b-mos.